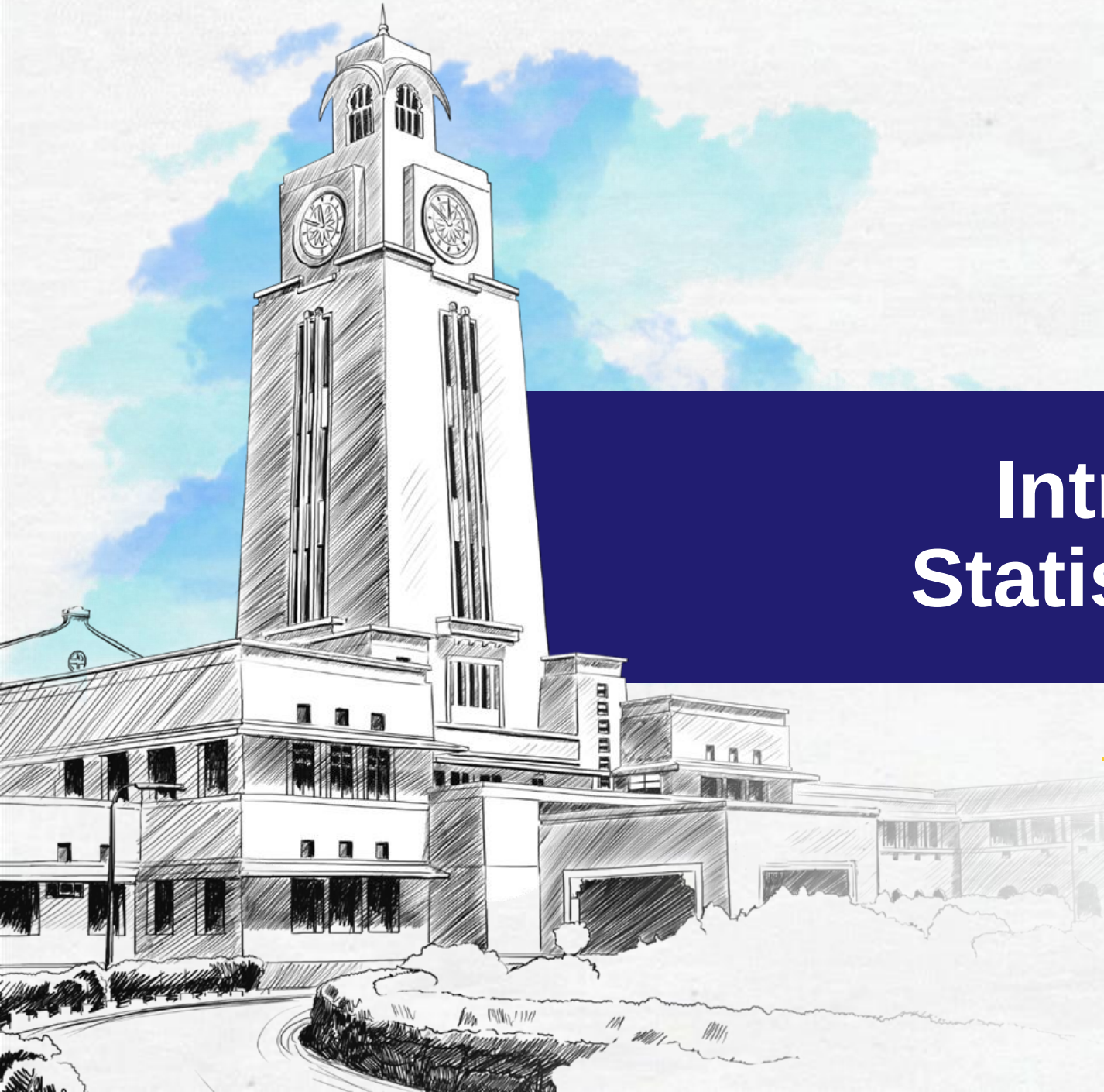


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Introduction to Statistical Methods

ISM Team

Section:
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LF:
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Session 7

Sampling and Distribution

Contact Session 7

CS - 7	Sampling – random sampling and Stratified sampling,Sampling distribution – Central Limit theorem,Estimation– Interval Estimation,Confidence level	T1 & T2
HW	Problems on Interval estimation	T1 & T2

Agenda

- Random Sampling
- Sampling Distribution and Central Limit Theorem,
- Point Estimation
- Interval Estimation
- Confidence Interval



Population and Sample

- **POPULATION** (Big data or Huge or Massive data)

A population is the aggregate of facts (Data). It is the set of all elements of interest for a specific Aim and Objectives. The process of conducting a survey to collect data for the entire population is called a census.

- **SAMPLE** (Small data)

A sample is a subset of the population. The process of conducting a survey to collect data for a sample is called a sample survey.

Basic Terminology

- Statistics uses data from a **sample** to make **estimates** and
- Testing of hypotheses about the characteristics of a population in the point of view of minimising the resources (Money, Manpower, Material and Time – **3MT**).
- This process of Statistics is called **Inferential statistics**.

Parameter and Statistic

➢ **Parameter:**

- It is population constants, whose values generally unknown
- The various statistical characteristics or constants of a population are:
  - Population Mean (μ), Population Variance (σ^2), Population Proportion (P), Population Correlation Coefficient (ρ) etc.

❑ Example: Binomial distribution (n, p)

Poisson distribution (λ)

Normal distribution (μ, σ^2)

Parameter and Statistic

- **Statistic**:

  A function of sample observations

- Example

Sample mean ($\bar{x}$), Sample variance (S^2), Sample proportion (P) etc.

- A **Statistic** also called an **Estimator** is used to estimate its corresponding **Parameter**. The numerical value is called estimate
- A statistic computed for different random samples forms a random variable and its distribution is called **Sampling distribution**.

Sampling

- Sampling is a process of selecting samples from a well-defined **target** populations. A target population is a population characteristic in which the Analyst is interested. For instance, the coping problem of all adolescents with elders.

- If the representative samples are selected from the well-defined target population which are also consistent and accurate, then the estimates obtained from such samples help in better generalization of the unknown parameters. This process is known as Inferential Statistics which forms a good scientific decision-making tool.

Reasons for Sampling

- Most often the population under study may be either unknown or infinite. Collecting data from such population is not feasible. Further, even if the population is countably finite, the resources required to collect data may be huge. Hence, by selecting representative samples from the target population, not only the resources can be minimized but also a close estimation of parameters can be made.

Continued…

- Collecting data from samples instead of population offers several Advantages:

1. The sample can save money.

2. The sample can save time.

3. For given resources, the sample can broaden the scope of the study.

Methods of Selection of Sample

- The data obtained from the samples are the observable data

Observed data = Truth + Bias + Random error

Sample design

Probability sampling

Procedure that assures all the units in the population have some **probabilities known in advance** of being chosen in a sample

Non-probability sampling

Procedures in which units in the sample are collected with no specific probability structure

Methods for Selection of Sample

- Probability sampling
  - Simple Random Sampling
  - Systematic Random Sampling
  - Multistage Random Sampling
  - Stratified Random Sampling

- Non – Probability sampling
  - Purposive / Judgement Sampling
  - Convenience Sampling
  - Quota Sampling
  - Snowball Sampling

Methods of Selection of Sample

- **The requirement for simple random sampling is**
  - Homogeneous and finite target population
  - Samples are selected unit by unit with equal probability
- **The requirement for systematic random sampling is**
  - Sampling frame (Listing of sampling units from finite population)
  - All units must be arranged in a systematic order (The order may be w.r.t alphabets, date of births, Adhaar numbers, telephone numbers.. etc)
  - Only the first unit is selected at random for the sample
  - Remaining units are selected at a regular interval of k (k = N/n)

Stratified Random Sampling

- **The requirement for Stratified random sampling is**
  - Heterogeneous target population
  - Divide the heterogeneous target population into different stratum by ensuring homogeneity within each stratum
  - Use simple random sampling within each stratum to select samples proportional to population size
  - For example, if there are three stratum with respective population sizes 2000, 6000, and 1000. The sample size is 1500, then with probability proportional to population size (PPPS) the respective samples are 333, 1000 and 167. (eg., 2000x1500/9000=333)
  - Obtain the estimates from each stratum and compute an estimate by weighted average.

Sampling Error

- which occurs when the sample is not a representative of the population.
- Improper sampling techniques used for selecting the samples.
- Sampling error is proportionate to sampling variation.

Sampling Variation

Population of Wages of employees of an organization

1861	2495	1000	2497	1865	791	2090	2637	1327	1678
1680	2858	795	2495	2496	2501	1160	1480	1860	2490
2090	2840	2490	2640	659	827	2646	2638	2643	868
1327	1866	1861	2486	2865	3011	2494	1489	1865	2855
2840	2499	2093	2660	1165	2600	2085	2640	2998	1861
2956	2495	2865	1865	3000	3019	1670	2858	2642	1680
3038	3000	1313	596	656	3240	590	2501	2485	3015
2092	1679	3024	2497	2825	2630	2070	2900	1861	2636
2495	2637	2497	1159	2640	3050	870	2896	2500	2638
926	2860	1481	875	2482	1860	2086	934	3200	2490

Selecting Samples of Different Sizes

Sample 1											
3000	2486	820	1678	2070	2638	2490	1865	1000	2090	596	3200
Sample 2											
2840	2858	3000	2490	2998	3050	2070	2896	3200	2490	3280	
Sample 3											
2858	3240	2497	2865	656	2093	934	1861	868	795		
Sample 4											
2086	1000	2497	596	656	875	2085	934	1313			
Sample 5											
820	1313	3000	2640	596	2640	2600	2495	934	2500		

Continued….

Sample 6										
2840	2499	1327	1861	2495	3024	3038	2497			
Sample 7										
2858	2490	868	1670	1480	2643	1480	1680	2085	2490	
Sample 8										
2495	2858	1861	2092	2499	3000	2660	1000	1679	926	2660
Sample 9										
795	791	3200	2085	2638	2497	2486	1159	2640		
Sample 10										
3019	3240	3200	3050	3000	3015	2900	2896	2998		

Calculation of Mean and Variance

Sample No.	Sample size	Mean	SD	Variance
1	12	1994.42	843.23	711036.83
2	11	2830.18	349.94	122458.00
3	10	1866.70	988.57	977270.64
4	9	1338.00	704.36	496123.01
5	10	1953.80	920.44	847209.79
6	8	2447.63	590.64	348855.61
7	10	1974.40	638.05	407107.80
8	11	2157.27	715.1	511368.01
9	9	2032.33	891.53	794825.74
10	9	3035.33	117.4	13782.76
Overall	100	**2162.24**	**732.26**	**536204.71**

Sampling Variation

- The term **"sampling variation"** refers to the fact that the sampling estimates vary from one sample to another forming a random variable.

- **Sampling variability** will **decrease** as the **sample size increases.**

- The samples must be randomly chosen. The size of the samples should be optimum (neither too small nor too large). The required optimum sample size are computed based on some statistical procedures like proportion, SD etc.

Sampling Distribution Definition

- The probability distribution of a statistic (sample estimate) is called **sampling distribution**.

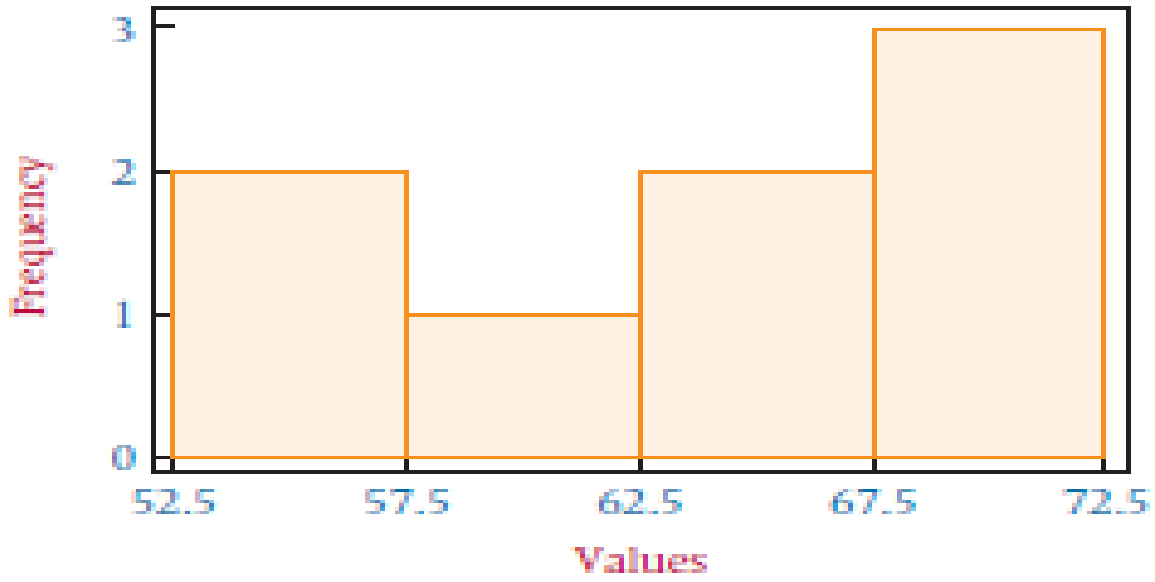
- The sampling distribution of a statistic depends on the distribution of the population, the size of the sample, and the method of sample selection

Sampling Distribution of Mean $\bar{x}$

- The sample mean is one of the more common statistic used in the inferential statistics.
- The **distribution** of the values of the sample mean ($\bar{x}$) in repeated **samples** is called the **sampling distribution of $\bar{x}$**
- One way to examine the distribution possibilities is to take a population with a particular distribution, randomly select samples of a given size, compute the sample means, and attempt to determine how the means are distributed.

Example

- Suppose a small finite population consists of only $N = 8$ numbers: 54, 55, 59, 63, 64, 68, 69, 70
- Using an Excel-produced histogram, we can see the shape of the distribution of this population of data.
- Suppose we take all possible samples of size $n = 2$ from this population with replacement.



Sample of Size 2

The result is the following pairs of data

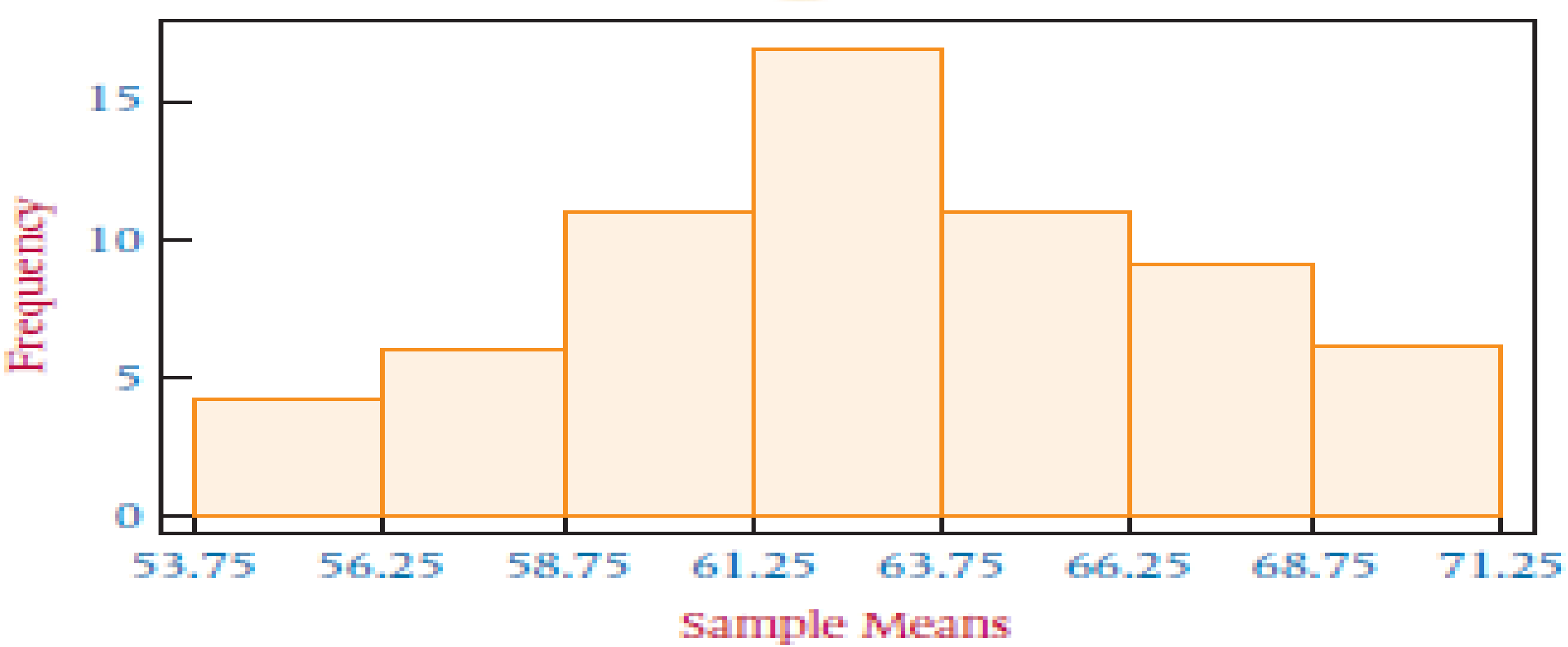
(54,54)	(55,54)	(59,54)	(63,54)
(54,55)	(55,55)	(59,55)	(63,55)
(54,59)	(55,59)	(59,59)	(63,59)
(54,63)	(55,63)	(59,63)	(63,63)
(54,64)	(55,64)	(59,64)	(63,64)
(54,68)	(55,68)	(59,68)	(63,68)
(54,69)	(55,69)	(59,69)	(63,69)
(54,70)	(55,70)	(59,70)	(63,70)
(64,54)	(68,54)	(59,54)	(70,54)
(64,55)	(68,55)	(69,55)	(70,55)
(64,59)	(68,59)	(69,59)	(70,59)
(64,63)	(68,63)	(69,63)	(70,63)
(64,64)	(68,64)	(69,64)	(70,64)
(64,68)	(68,68)	(69,68)	(70,68)
(64,69)	(68,69)	(69,69)	(70,69)
(64,70)	(68,70)	(69,70)	(70,70)

The means of each of these samples follow

54	54.5	56.5	58.5	59	61	61.5	62
54.5	55	57	59	59.5	61.5	62	62.5
56.5	57	59	61	61.5	63.5	64	64.5
58.5	59	61	63	63.5	65.5	66	66.5
59	59.5	61.5	63.5	64	66	66.5	67
59	61.5	63.5	65.5	66	68	68.5	69
61.5	62	64	66	66.5	68.5	69	69.5
62	62.5	64.5	66.5	67	69	69.5	70

Sampling Distribution

- Again, using an Excel-produced histogram, we can see the shape of the distribution of these sample means.



Conclusion

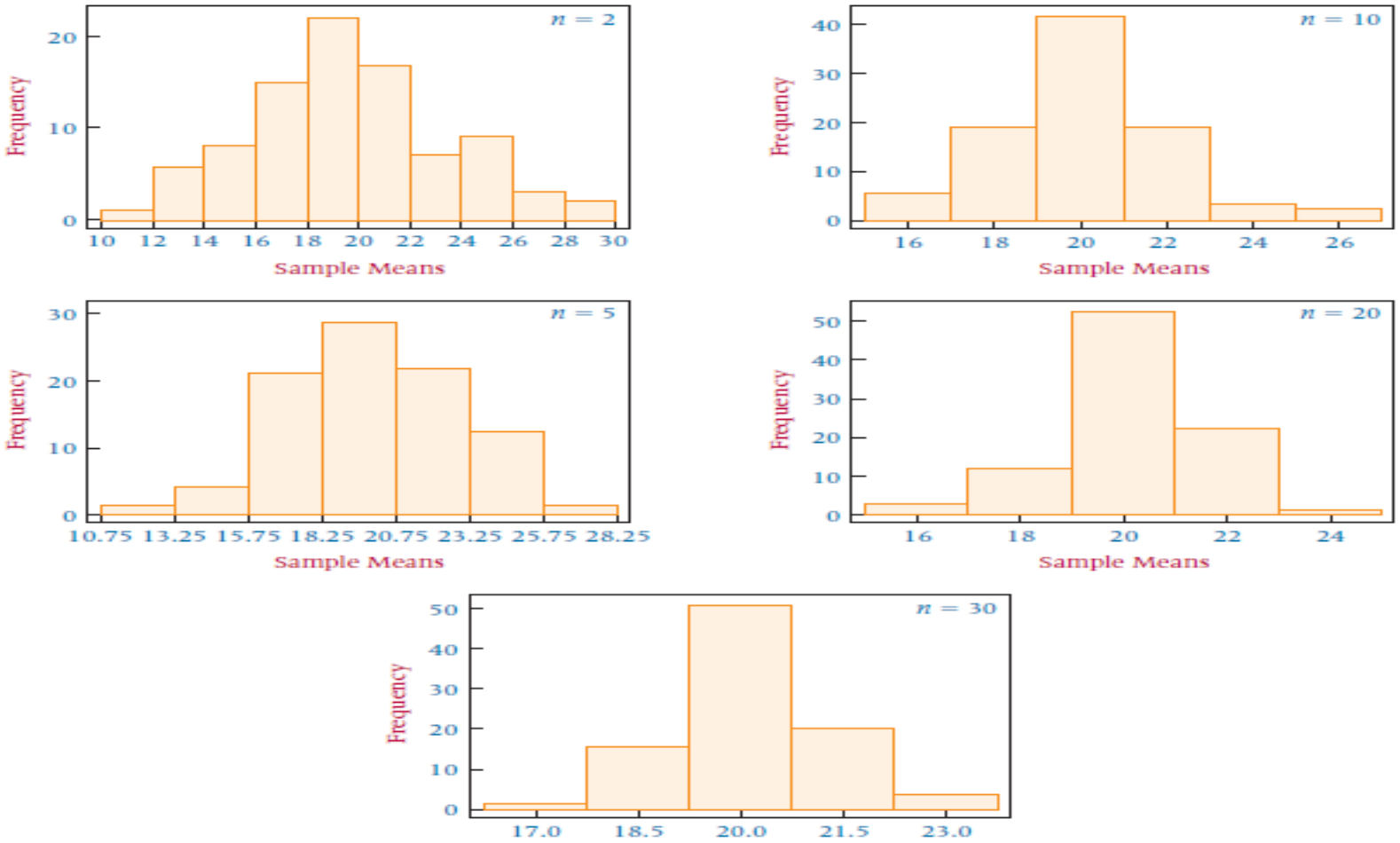
- Observe that the shape of the histogram for sample means is quite unlike the shape of the histogram for the population.

- The sample means appear to “pile up” toward the middle of the distribution and “tail off” toward the extremes.

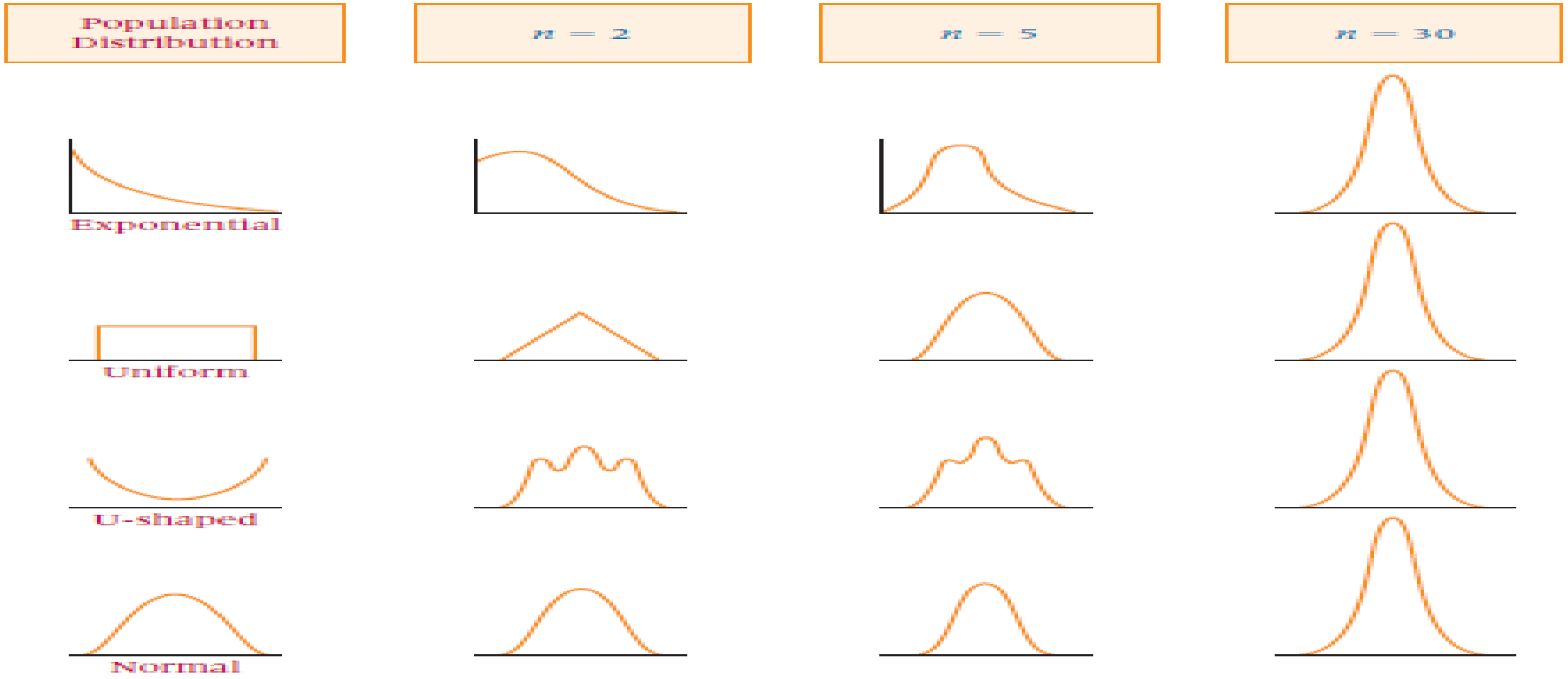
- As sample sizes become much larger, the sample mean distributions begin to approach a **Normal distribution** and the variation among the means decreases.

Sampling Disrtibution Analysis

- Sample Means from 90 Samples Ranging in Size from
- n = 2 to n = 30 from a Uniformly Distributed Population with a = 10 and b = 30



Shapes of the Distributions of Sample Means



Central Limit Theorem

- If samples of size *n* are drawn randomly from a population that has a mean of μ and a standard deviation of σ, the sample means, $\bar{x}$, are approximately normally distributed for sufficiently large sample sizes ($n \geq 30$) regardless of the shape of the population distribution.

- If the population is normally distributed, the sample means are normally distributed for any size sample.

- From mathematical expectation: $\mu_{\bar{x}} = \mu$ and $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

Z score for sample means

- The central limit theorem states that sample means are normally distributed regardless of the shape of the population for large samples and for any sample size with normally distributed populations.
- Thus, **sample means** can be **analyzed** by using ***z* scores**
- The formula to determine z scores for individual values from a normal distribution: $Z = \frac{X-\mu}{\sigma}$
- If sample means are normally distributed, the z score formula applied to sample means would be $Z = \frac{\bar{X}-\mu_{\bar{X}}}{\sigma_{\bar{X}}}$
- The standard deviation of the statistic of interest is $\sigma_{\bar{X}}$, sometimes referred to as the **standard error of the mean.**

Example-1

The average age of a vehicle registered in the United States is 8 years, or 96 months. Assume the standard deviation is 16 months. If a random sample of 36 vehicles is selected, find the probability that the **mean** of their age is between 90 and 100 months.

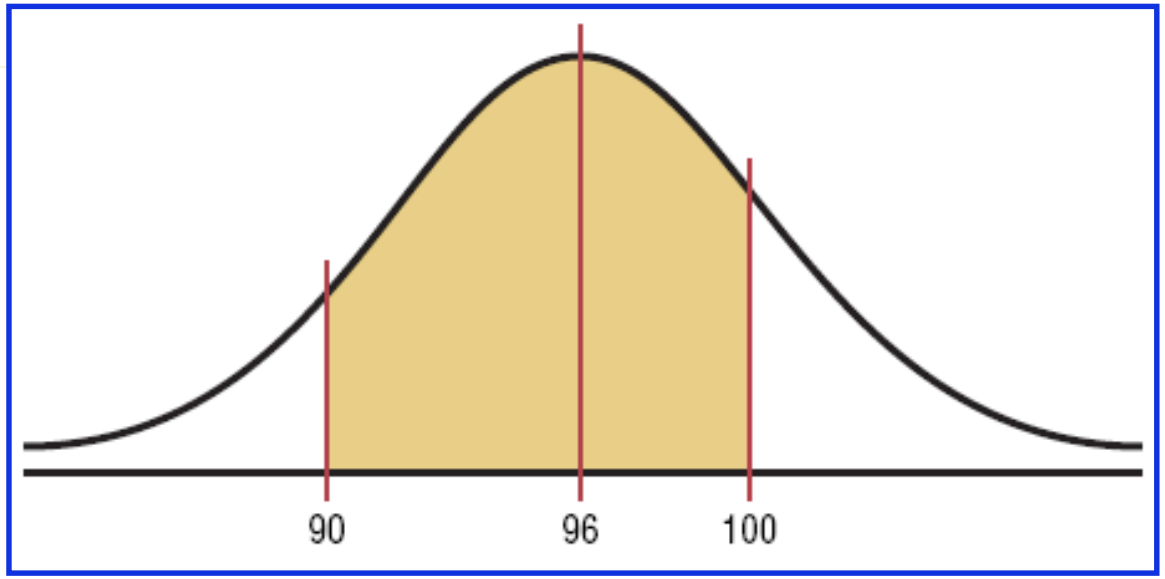
Given: $\mu = 96$ months, & $\sigma = 16$, $n = 36 > 30 \rightarrow CLT$

$$z = \frac{x - \mu}{\sigma}, z = \frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$z = \frac{90 - 96}{\frac{16}{\sqrt{36}}} = -2.25 \qquad z = \frac{100 - 96}{\frac{16}{\sqrt{36}}} = 1.50$$

$$P(90 < \bar{x} < 100) = \quad P(-2.25 < z < 1.5)$$

$$= 0.9332 - 0.0122 = 0.9210$$



Example-2

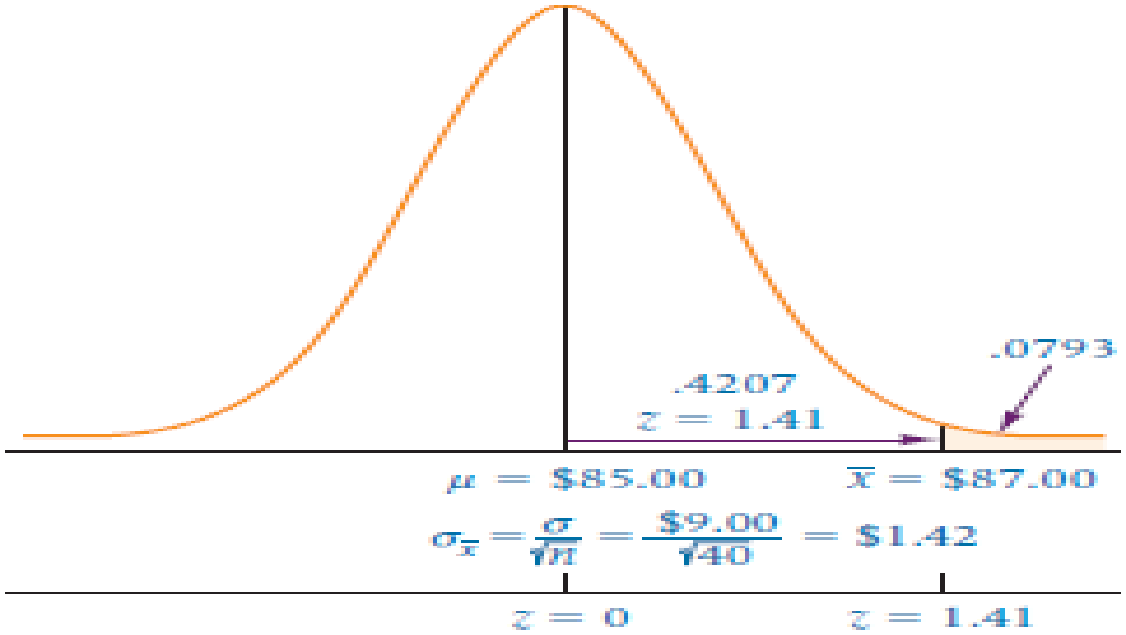
Suppose the mean expenditure per customer at a tire store is $85.00, with a standard deviation of $9.00. If a random sample of 40 customers is taken, what is the probability that the sample average expenditure per customer for this sample will be $87.00 or more?

Solution: Because the sample size is greater than 30, the central limit theorem

Can be used, and the sample means are normally distributed.

μ= $85 σ= $9

$$z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} = \frac{\$87.00-\$85.00}{\frac{\$9.00}{\sqrt{40}}} = \frac{\$2.00}{\$1.42} = 1.4$$



STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91309	.91466	.91621	.91774
1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
2.0	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361

Example-3

Suppose that during any hour in a large department store, the average number of shoppers is 448, with a standard deviation of 21 shoppers. What is the probability that a random sample of 49 different shopping hours will yield a sample mean between 441 and 446 shoppers?

For this problem, μ= 448, σ= 21, and n = 49. The problem is to determine P(441<= $\bar{x}$ <=446).

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$z_1 = \frac{441-448}{\frac{21}{\sqrt{49}}} = \frac{-7}{3} = -2.33\ , \quad z_2 = \frac{446-448}{\frac{21}{\sqrt{49}}} = \frac{-2}{3} = -0.67$$

Now, P(-2.33<z<-0.67)

=P(0.67<z<2.33)

=F(2.33)-F(0.67)

=0.9901-0.74851

=0.24159

Finite Population Sampling

The earlier example was based on the assumption that the population was infinitely or extremely large.

In cases of a finite population, *a statistical adjustment can be made to the z formula for sample means*. The adjustment is called the **finite correction factor** $\sqrt{\frac{N-n}{N-1}}$

Following is the z formula for sample means when samples are drawn from finite populations. $z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}}$

Note about Finite Population

- As the size of the finite population becomes larger in relation to sample size, the finite correction factor approaches 1.

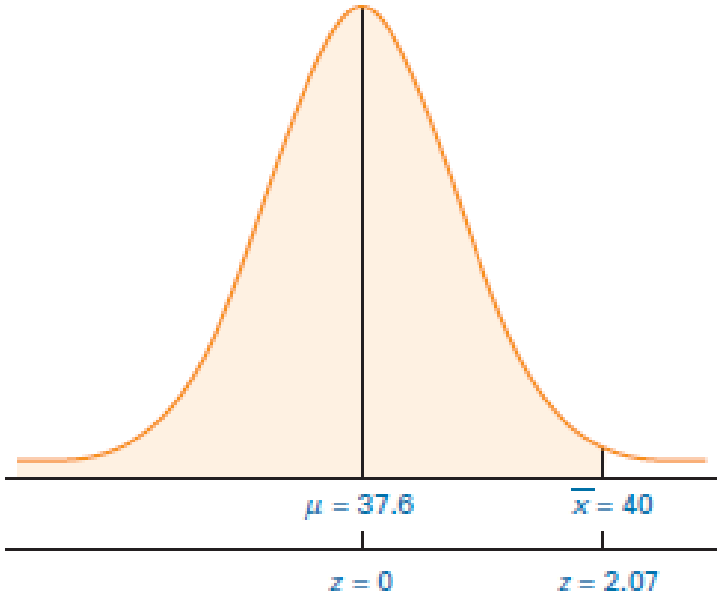
- In theory, whenever researchers are working with a finite population, they can use the finite correction factor.

- A rough rule of thumb for many researchers is that, if the sample size is **less** than **5%** of the finite population size or $n/N < 0.05$, the finite correction factor does **not** significantly modify the solution.

Example-4

A production company's 350 hourly employees average 37.6 years of age, with a standard deviation of 8.3 years. If a random sample of 45 hourly employees is taken, what is the probability that the sample will have an average age of less than 40 years?

Solution: The population mean is 37.6, with a population standard deviation of 8.3. The sample size is 45, but it is being drawn from a finite population of 350; that is, n = 45 and N = 350. The sample mean under consideration is 40. Using the z formula with the finite correction factor gives



$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}} = \frac{40 - 37.6}{\frac{8.3}{\sqrt{45}}\sqrt{\frac{350-45}{350-1}}} = \frac{2.4}{1.157} = 2.07$$

Thus, P(z<2.07) = 0.9808 is the required probability.

Sampling Distribution of Sample Proportion

If research results in ***countable*** items such as how many elements satisfy a certain property or characteristic, the **sample proportion** is often the statistic of choice.

SAMPLE PROPORTION

$$\hat{p} = \frac{x}{n}$$

Where

x= number of items in a sample that have the characteristic

n= number of items in the sample

Example

In a sample of 100 factory workers, 30 workers might belong to a union. The value of sample proportion for this characteristic, union membership, is 30/100 = 0 .30

Central Limit Theorem for Sample Proportion

- The central limit theorem applies to sample proportions is that the normal distribution approximates the shape of the distribution of sample proportions

  If *n*p* > *15* and *n*q* > *15* (*p* is the population proportion and *q* = 1 - *p*).

- The mean of sample proportions for all samples of size *n* randomly drawn from a population is ***p*** (the population proportion) and
- the standard deviation of sample proportions (sometimes referred to as the **standard error of the proportion) is given by**

Finite Population	*Infinite Population*
$\sigma_{\bar{p}} = \sqrt{\frac{N-n}{N-1}}\sqrt{\frac{p(1-p)}{n}}$	$\sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}}$

Z Formula for Sample Proportion

For $n*p > 15$ and $n*q > 15$

$$Z = \frac{\hat{p} - P}{\sqrt{\frac{pq}{n}}}$$

where

$P = Population\ proportion$

$\hat{p} = Sample\ proportion$

$q = 1-p$

n = Sample size

Example-4

Suppose 60% of cancer patient are cured by new drug. What is the probability that a random sample of size 120 patients 50% or more will cured by new drug?

Solution: Here $p = 0.6$, $\hat{P} = 0.5$, n = 120, $q = 0.4$

$$Z = \frac{\hat{p} - P}{\sqrt{\frac{pq}{n}}} = \frac{0.5 - 0.6}{\sqrt{\frac{0.5 * 0.5}{120}}} = -2.24$$

Therefore, P(z > -2.24)

= 1- P(z<-2.24)

= 1- 0.**0125 = 0.9875**

Statistical Inference

Three forms of statistical inference

- Point estimation
- Interval estimation
- Hypothesis testing

Point Estimate

- A **point estimate** is a statistic taken from a sample that is used to estimate a population parameter.
- A point estimate is only as good as the representativeness of its sample.
- If other random samples are taken from the population, the point estimates derived from those samples are likely to vary.

Terminology and Notations

Statistical measure	Sample estimates	Population parameter
Mean	$\overline{x}$	μ
Variance	s^2	σ^2
Proportion	$\overline{p}$	P
Correlation	r	ρ

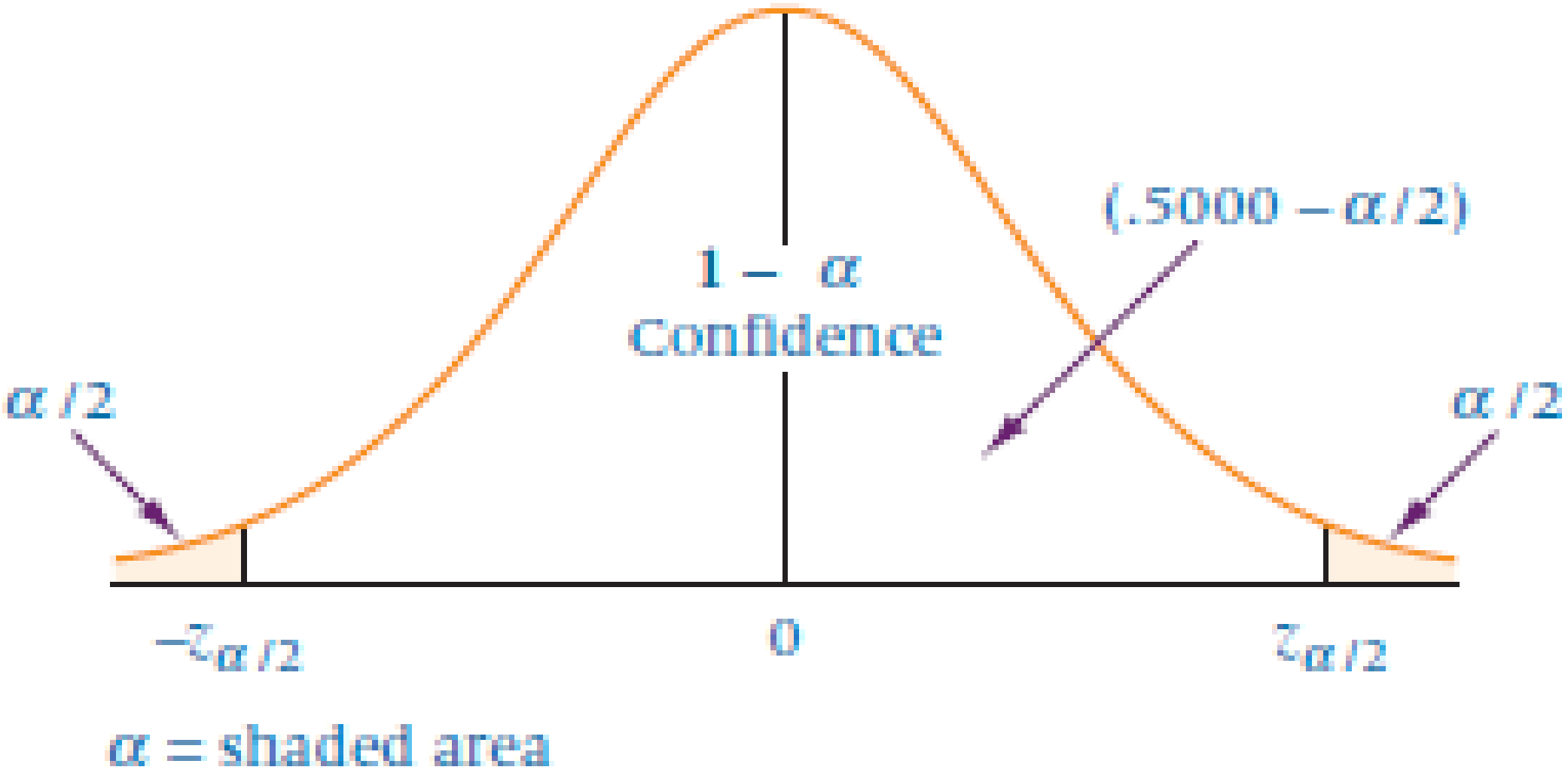
Confidence Interval

- Computation of 100 (1-α)% confidence interval is the most common way of finding the interval estimate, where α is the probability of type I error. α is also called area under curve at one of the tail-end of normal distribution. $\frac{\alpha}{2}$ is the area under normal curve at both the tail-ends.

- Most commonly used confidence level close to 1, are 0.95 (95%) or 0.99 (99%).

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Graphical Representation of Confidence Interval



α in Confidence Intervals

Standard normal variable value $z_{\alpha/2}$ (Table Value)	Level of significance $\propto$		
	1%=0.01	**5%=0.05**	**10%=0.1**
$\propto$/2	0.01/2 =0.005	0.05/2=0.025	0.1/2=0.05
$F(z_{\alpha/2})=1-\propto/2$ $P(Z<=z_{\alpha/2})= 1-\propto/2$	P(Z≤2.58)= 0.995 then $z_{\alpha/2}= 2.58$	P(Z≤1.96)= 0.975 then $z_{\alpha/2}= 1.96$	P(Z≤1.645)= 0.95 then $z_{\alpha/2}= 1.645$

Confidence Interval For Mean

The 100 (1-α)% confidence interval for mean is

$$\text{Sample estimate} \pm Z_{\alpha/2}\,SE\,(\text{estimate})$$

Based on Standard normal distribution 100 (1-α)% CI for μ is

$$\bar{x} \pm Z_{\alpha/2}\,SE\,(\bar{x})$$

For α = 0.05, $Z_{\alpha/2} = 1.96$

For α = 0.01, $Z_{\alpha/2} = 2.58$

Confidence Interval for Difference of Mean of Two Populations

The 100 (1-α)% confidence interval for difference in means of two populations is

$$\text{Difference in estimates} \pm Z_{\frac{\alpha}{2}}\, SE(Difference\ in\ \text{estimates})$$

Based on Standard normal distribution 100 (1-α)% CI is

$$(\bar{X}_1 - \bar{X}_2) \pm Z_{\frac{\alpha}{2}}\, SE(\bar{X}_1 - \bar{X}_2)$$

For α = 0.05, $Z_{\alpha/2} = 1.96$

For α = 0.01, $Z_{\alpha/2} = 2.58$

$$SE\,(\bar{x}_1 - \bar{x}_2) = \sqrt{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)}$$

Example

In the cellular telephone company, problem of estimating the population mean number of minutes called per residential user per month, from the sample of 85 bills it was determined that the sample mean is 510 minutes. Suppose past history and similar studies indicate that the population standard deviation is 46 minutes.Determine a 95% confidence interval.

The business researcher can now complete the cellular telephone problem. To determine a 95% confidence interval for $\bar{x}$=510, $\sigma = 46$, n=85 , and z=1.96, the researcher estimates the average call length by including the value of z in formula 8.1.

$$510\text{-}1.96\frac{46}{\sqrt{85}} \leq \mu \leq 510 + 1.96\frac{46}{\sqrt{85}}$$

$$510\text{-}9.78 \leq \mu \leq 510+9.78$$

$$500.22 \leq \mu \leq 519.78$$

Example

A study is conducted in a company that employs 800 engineers. A random sample of 50 engineers reveals that the average sample age is 34.3 years. Historically, the population standard deviation of the age of the company's engineers is approximately 8 years. Construct a 98% confidence interval to estimate the average age of all the engineers in this company.

- This problem has a finite population. The sample size, 50, is greater than 5% of the population, so the finite correction factor may be helpful.
- In this case N = 800, n = 50, $\bar{x}$= 34.3 and σ= 8
- The z value for a 98% confidence interval is 2.33

$$\bar{x} - z\alpha/_2 \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \leq \mu \leq \bar{x} - z\alpha/_2 \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

$$34.30\text{-}2.33\frac{8}{\sqrt{50}} \sqrt{\frac{800-50}{800-1}} \leq \mu \leq 34.30\text{+}2.33\frac{8}{\sqrt{50}} \sqrt{\frac{800-50}{800-1}}$$

$$34.30\text{-}2.33\frac{8}{\sqrt{50}} \sqrt{\frac{750}{799}} \leq \mu \leq 34.30\text{+}2.33\frac{8}{\sqrt{50}} \sqrt{\frac{750}{799}}$$

$$34.30\text{-}2.55 \leq \mu \leq 34.30\text{+}2.55$$

$$31.75 \leq \mu \leq 36.85$$

Example

A survey was taken of U.S. companies that do business with firms in India. One of the questions on the survey was: Approximately how many years has your company been trading with firms in India? A random sample of 44 responses to this question yielded a mean of 10.455 years. Suppose the population standard deviation for this question is 7.7 years. Using this information, construct a 90% confidence interval for the mean number of years that a company has been trading in India for the population of U.S. companies trading with firms in India.

Solution

Here, n= 44, $\bar{x}$= 10.455 and σ= 7.7. To determine the value of $z_{\alpha/2}$, divide the 90% confidence in half, or take .5000 - α/2= .5000 - .0500= 0.45 where α= 10%.

Z table yields a z value of 1.645 for the area of .45

The confidence interval is

$$x - z\frac{\sigma}{\sqrt{n}} \leq \mu \leq x + z\frac{\sigma}{\sqrt{n}}$$

$$10.455\text{-}1.645\left(\frac{7.7}{\sqrt{44}}\right) \leq \mu \leq 10.455\text{+}1.645\left(\frac{7.7}{\sqrt{44}}\right)$$

$$10.455\text{-}1.910 \leq \mu \leq 10.455\text{+}1.910$$

$$8.545 \leq \mu \leq 12.365$$

Practice Problem-1

In an effort to estimate the mean amount spent per customer for dinner at a major Atlanta restaurant, data were collected for a sample of 49 customers. Assume a population standard deviation of $5. a. At 95% confidence, what is the margin of error? b. If the sample mean is $24.80, what is the 95% confidence interval for the population mean?

Solution:

a. $1.96\sigma/\sqrt{n} = 1.96(5/\sqrt{49}) = 1.40$

b. 24.80 ± 1.40; 23.40 to 26.20

Practics Problem-2

The amount of time a bank teller spends with each customer has a population mean, of 3.10 minutes and population standard deviation, of 0.40 minute. If you select a random sample of 16 customers,

a. what is the probability that the mean time spent per customer is at least 3 minutes?

b. there is an 85% chance that the sample mean is less than how many minutes?

c. If you select a random sample of 64 customers, there is an 85% chance that the sample mean is less than how many minutes?

Solution:

(a) $P(\bar{X} > 3) = P(Z > -1.00) = 1.0 - 0.1587 = 0.8413.$

(b) $P(Z < 1.04) = 0.85 \quad \bar{X} = 3.10 + 1.04(0.1) = 3.204.$

c) $P(Z < 1.04) = 0.85, \bar{X} = 3.10 + 1.04(0.05) = 3.152$

Practice Problem-3

A political pollster is conducting an analysis of sample results in order to make predictions on election night. Assuming a two-candidate election, if a specific candidate receives at least 55% of the vote in the sample, then that candidate will be forecast as the winner of the election. If you select a random sample of 100 voters, what is the probability that a candidate will be forecast as the winner when

a. the true percentage of her vote is 50.1%?

b. the true percentage of her vote is 60%?

c. the true percentage of her vote is 49% (and she will lose the election)?

d. If the sample size is increased to 400, what are your answers to (a) through (c)?

Solution

(a) $\pi = 0.501$, $\sigma_P = \sqrt{\frac{\pi(1-\pi)}{n}} = \sqrt{\frac{0.501(1-0.501)}{100}} = 0.05$

$P(p > 0.55) = P(Z > 0.98) = 1.0 - 0.8365 = 0.1635$

(b) $\pi = 0.60$, $\sigma_P = \sqrt{\frac{\pi(1-\pi)}{n}} = \sqrt{\frac{0.6(1-0.6)}{100}} = 0.04899$

$P(p > 0.55) = P(Z > -1.021) = 1.0 - 0.1539 = 0.8461$

(c) $\pi = 0.49$, $\sigma_P = \sqrt{\frac{\pi(1-\pi)}{n}} = \sqrt{\frac{0.49(1-0.49)}{100}} = 0.05$

$P(p > 0.55) = P(Z > 1.20) = 1.0 - 0.8849 = 0.1151.$

(d) Increasing the sample size by a factor of 4 decreases the standard error by a factor of 2.

Practics Problem-4

The following data represent the bounced cheque fee, in dollars, charged by a sample of 23 banks for direct-deposit customers who maintain a $100 balance:

26, 28, 20, 20, 21, 22, 25, 25, 18, 25, 15, 20, 18, 20, 25, 25, 22, 30, 30, 30, 15, 20, 29

a.Construct a 95% confidence interval for the population mean bounced cheque fee.

b.Interpret the interval constructed in (a)

Solution:

(a) $21.01 to $24.99.

(b) You can be 95% confident that the population mean bounced-cheque fee is between $21.01 and $24.99

Practice Problem-5

- A survey is planned to determine the mean annual family medical expenses of employees of a large company. The management of the company wishes to be 95% confident that the sample mean is correct to within $50 of the population mean annual family medical expenses. A previous study indicates that the standard deviation is approximately $400.

a. How large a sample size is necessary?

b. If management wants to be correct to within $25, what sample size is necessary?

Solution:

$$\textbf{(a)}\ n = \frac{Z^2\sigma^2}{e^2} = \frac{(1.96^2)(400^2)}{50^2} = 245.86$$

Use $n = 246$.

$$\textbf{(b)}\ n = \frac{Z^2\sigma^2}{e^2} = \frac{(1.96^2)(400^2)}{25^2} = 983.41$$

Use $n = 984$.

Glossary

- Population
- Sample
- Random sampling
- Stratified sampling
- Parameter
- Statistics
- Sampling distribution

- Point estimate
- Interval estimation
- Confidence level
- Confidence interval

Thank You!